Factors to consider when selecting simulation package

1. **General Capabilities:**
    - Modelling Flexibility: The software should allow modelling of systems with varying levels of complexity. It should support the definition and modification of attributes for entities and global variables, as well as mathematical expressions and functions.
    - Ease of Use: The software should be user-friendly and have modelling constructs that strike a balance between simplicity and flexibility. Graphical user interfaces, hierarchical modelling, and debugging aids enhance usability.
    - Debugging Aids: Interactive debugging tools allow users to track entities, observe model states, and manipulate attributes for testing and validation.
    - Execution Speed: Faster execution speeds are desirable, especially for large models or applications requiring high processing rates.
    - User-Friendly Interfaces: Ability to create user-friendly interfaces for non-expert users, facilitating easy input of model parameters.
    - Run-Time Versions: Providing run-time versions allows users to modify model data without altering the logic, enabling various applications such as training or sales demonstrations at a lower cost.
    - Data Import and Export: Capability to import and export data from/to other applications like spreadsheets or databases enhances interoperability.
    - Cost Modules: Some software includes cost modules for assigning costs to different components of the simulated system.
    - Combined Discrete-Continuous Simulation: For certain applications, the ability to integrate discrete-event and continuous simulation capabilities is necessary.
    - Integration with External Routines: Integration with external routines allows complex logic written in programming languages to be incorporated into the simulation model.

- Initialization and State Saving: Easy initialization of the simulation in a desired state and the ability to save and restart simulations are useful features.

2. **Hardware**
    - Ensure compatibility with existing hardware configurations.
    - Check for any additional software requirements or dependencies.

3. **Animation**
    - Uses of Animation:
        i. Communicating Model Essence: Animation helps convey the essence of the simulation model to stakeholders who may not be familiar with technical details.
        ii. Debugging: It aids in debugging the simulation program by visually representing the model's behaviour.
        iii. Model Validity: Can demonstrate if a simulation model is valid or identify areas where it may need improvement.
        iv. Operational Procedures: Helps suggest improvements in operational procedures by visually stimulating different scenarios.
        v. Training: Useful for training operational personnel by visually illustrating system behaviour.
        vi. Team Communication: Promotes communication among project teams by providing a visual representation of the system.
    - Types of Animation:
        i. Concurrent Animation: Displayed while the simulation is running, allowing real-time visualisation of the model's behaviour. However, it's usually turned off during production runs due to performance considerations.

ii. Post-Processed Animation: State changes in the simulation are saved to a file and used to generate graphics after the simulation ends.

- Desirable Features:
  i. Simulation software should provide default animation capabilities as part of the model-building process.
  ii. Ability to create and save high-resolution icons for reuse.
  iii. Icons should move smoothly without flashing or jumping.
  iv. Users should be able to control animation speed, zoom in/out, and pan to view different parts of the system.
  v. Allow users to create and save different views corresponding to various parts of the system - Named animation views
  vi. Vector-Based graphics preferable over pixel-based graphics for better scalability and rotation capabilities.
  vii. Some software allows users to pause the simulation, make parameter changes, and resume simulation.
  viii. Provide the ability to view the animation from different angles, which can be useful for management presentations.
  ix. Capability to import CAD drawings and clip art for use in animation.
  x. Dynamic Graphics: Ability to display dynamic graphics and statistics on the screen during simulation execution, such as clocks, dials, and dynamically updated plots.

4. Statistical capabilities

- If a simulation program lacks good statistical features, it's hard to trust its results.
- First, it needs a reliable random number generator to create realistic scenarios. Then, it should let us run the simulation multiple times with different starting points to see how things change.

- It should also help us figure out how confident we can be in our results and allow us to choose what data we want to analyse.
- Lastly, some software can even help us find the best solutions to problems by trying out different options automatically.

5. **Support and documentation**
    - Comprehensive documentation is essential for users to understand and utilise the software effectively. This includes user guides, detailed examples, and context-dependent online help.
    - A detailed description of each modelling construct should be provided, especially for complex procedures.
    - Having a high-quality textbook available can greatly aid users in learning and mastering the software.
    - Many vendors offer free demos or downloadable versions of their software for users to test out before purchasing.
    - Vendors should consider publishing electronic newsletters and hosting annual user conferences to keep users informed and engaged.
    - Regular software updates are also important to ensure that users have access to the latest features and improvements.

6. **Reports and Graphics**
    - Standard reports should be available for estimated performance measures, but customization options for reports, especially for management presentations, should also be possible.
    - Users should be able to compute estimates of user-defined performance measures and include them in custom reports.
    - For each performance measure, such as time in the system, average, minimum, and maximum observed values are typically provided. However, caution should be exercised with standard deviation values, ensuring they are based on statistically acceptable methods.

- Reports should be obtainable at intermediate points during a simulation run, not just at the end.
- The software should offer various static graphics capabilities.
    i. Histograms are useful for visualising sets of observed data, providing graphical estimates of underlying probability functions.
    ii. Time plots show key system variables over the simulation length, offering insights into dynamic system behaviour.
    iii. Bar charts, pie charts, and correlation plots are also valuable visualisation tools.
- Additionally, the ability to export individual model output observations to other software packages for further analysis and display, such as spreadsheets, databases, and statistical packages, should be supported by the simulation software.